

Let  $x \neq 2m\pi$  ( $m \in \mathbb{Z}$ ) be a real number. Observe that

$$\begin{aligned} \sum_{k=1}^n e^{ikx} &= e^{ix} \cdot \sum_{k=0}^{n-1} e^{ikx} = e^{ix} \cdot \frac{1 - e^{inx}}{1 - e^{ix}} = e^{ix} \cdot \frac{e^{\frac{inx}{2}}}{e^{\frac{ix}{2}}} \cdot \frac{e^{\frac{inx}{2}} - e^{-\frac{inx}{2}}}{e^{\frac{ix}{2}} - e^{-\frac{ix}{2}}} \\ &= e^{\frac{ix}{2} \cdot (n+1)} \cdot \frac{\sin \frac{nx}{2}}{\sin \frac{x}{2}} \end{aligned}$$

Since  $e^{ikx} = \cos kx + i \cdot \sin kx$  and

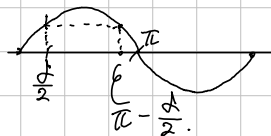
$$e^{\frac{ix}{2} \cdot (n+1)} = \cos \frac{(n+1)x}{2} + i \cdot \sin \frac{(n+1)x}{2}$$

So therefore

$$\sum_{k=1}^n \cos kx = \frac{\sin \frac{nx}{2}}{\sin \frac{x}{2}} \cdot \cos \frac{(n+1)x}{2} \quad \text{and} \quad \sum_{k=1}^n \sin kx = \frac{\sin \frac{nx}{2}}{\sin \frac{x}{2}} \cdot \sin \frac{(n+1)x}{2}$$

Now, for fixed  $0 < \delta < \pi$ , given  $x \in [\delta, 2\pi - \delta]$ ,

$$\left| \sum_{k=1}^n \cos kx \right|, \left| \sum_{k=1}^n \sin kx \right| \leq \left| \frac{1}{\sin \frac{x}{2}} \right| \leq \left| \frac{1}{\sin \frac{\delta}{2}} \right|$$



Consequently, we obtain a Theorem:

[Thm. Suppose that  $\{a_n\}$  converges to 0, and  $\sum |a_n - a_{n-1}|$  converges.

Given  $\delta \in (0, \pi)$ ,  $\sum_{n=1}^{\infty} a_n \sin nx$  and  $\sum_{n=1}^{\infty} a_n \cos nx$  converges uniformly on  $[\delta, 2\pi - \delta]$ .

Proof. Since the observation above,

$$\left| \sum_{n=1}^m \sin nx \right| \leq \left| \frac{1}{\sin \frac{\delta}{2}} \right| \quad \text{for all } \delta \leq x \leq 2\pi - \delta.$$

for all  $m \in \mathbb{N}$ , so  $\sum \sin nx$  is bounded uniformly. So, it satisfies the condition for the Dirichlet-DePold Test, the given series converges uniformly on  $[\delta, 2\pi - \delta]$ .

Thm. Let  $\{a_n\}$  be a monotone-decreasing real-numbers sequence, converging to 0. Then,  $\sum_{n=1}^{\infty} a_n \cdot \sin nx$  converges uniformly on  $\mathbb{R}$  if and only if  $\lim_{n \rightarrow \infty} n \cdot a_n = 0$ .

Proof. Suppose that  $\sum a_n \sin nx$  converges uniformly on  $\mathbb{R}$ .

By the Cauchy criterion, given  $\epsilon > 0$ , there is  $N \in \mathbb{N}$  such that

$$n > m \geq N \Rightarrow \left| \sum_{k=m}^n a_k \cdot \sin kx \right| < \epsilon \text{ for all } x \in \mathbb{R}.$$

Now, for any  $n \in \mathbb{N}$ , let  $m \in \mathbb{N}$  be the smallest integer bigger than  $\frac{n}{2}$ .

Then, for all  $k = m, m+1, \dots, n$ ,  $\frac{1}{2} \geq \frac{k}{2n} \geq \frac{m}{2n} \geq \frac{1}{4}$ . Therefore,

$$\sum_{k=m}^n a_n \cdot \sin \frac{k}{2n} \cdot \pi \stackrel{(1)}{\geq} \sum_{k=m}^n a_n \cdot \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} \cdot \sum_{k=m}^n a_n \stackrel{(2)}{\geq} \frac{1}{\sqrt{2}} \cdot \frac{n}{2} \cdot a_n$$

The (1) is given by the fact that  $\sin x$  increase on  $[0, \frac{\pi}{2}]$ .

The (2) is given by the fact that  $\{a_n\}$  decrease monotonically.

Now, the first term in the inequality tends to zero, so is  $n \cdot a_n$ .

Conversely, suppose that  $n a_n \rightarrow 0$  as  $n \rightarrow \infty$ . For each  $n \in \mathbb{N}$ , define  $E_n := \sup_{k \geq n} k \cdot a_k$ .

Let  $x \in [0, \pi]$  be fixed. Then, there exists a positive integer  $N \in \mathbb{N}$  such that

$$\frac{\pi}{N+1} < x < \frac{\pi}{N}$$

$$\text{So, } \left| \sum_{k=n}^{n+N-1} a_k \cdot \sin kx \right| \leq \sum_{k=n}^{n+N-1} a_k \cdot |\sin kx| \leq \sum_{k=n}^{n+N-1} a_k \cdot kx \leq N \cdot E_n \cdot x < \pi \cdot E_n$$

Meanwhile, let  $S_n = \sum_{k=1}^n \sin kx$ . Then, given  $M > n+N$ ,

$$\left| \sum_{k=n+N}^M a_k \cdot \sin kx \right| = \left| \sum_{k=n+N}^M a_k \cdot (S_k - S_{k-1}) \right|$$

$$= \left| \sum_{k=n+N}^M a_k \cdot S_k - \sum_{k=n+N}^M a_k \cdot S_{k-1} \right|$$

$$= \left| \sum_{k=n+N}^M [a_k - a_{k+1}] \cdot S_k + a_{M+1} \cdot S_M - a_{n+N} \cdot S_{n+N-1} \right|$$

$$\leq \left| \frac{1}{\sin \frac{x}{2}} \right| \cdot 2a_{n+N} \leq \frac{\pi}{x} \cdot 2a_{n+N} < 2(N+1) \cdot a_{n+N} < 2 \cdot (N+1) \cdot a_{n+N} < 2E_n$$

Consequently, for each  $n \in \mathbb{N}$ ,  $\left| \sum_{k=n}^{\infty} a_k \cdot \sin kx \right| < (2+\pi) \cdot E_n \rightarrow 0$  as  $n \rightarrow \infty$

So the given series converges uniformly on  $[0, \pi]$ , the periodicity gives convergence on  $\mathbb{R}$ .